2021

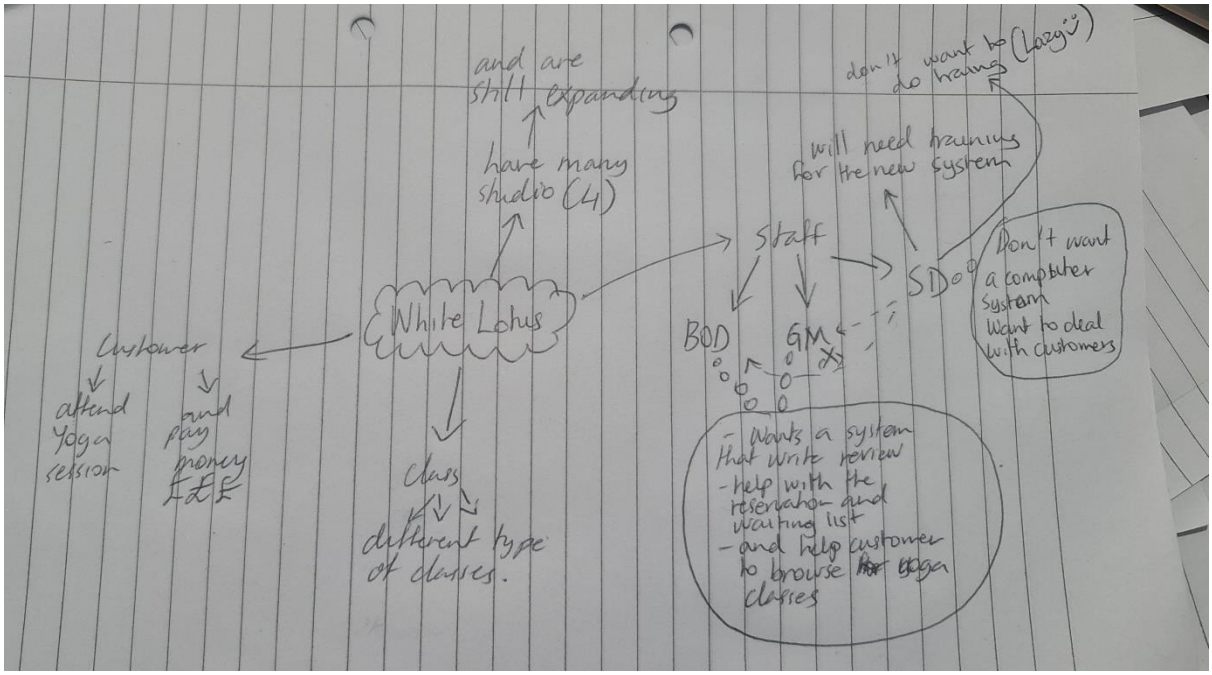
White Lotus

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Problem



This is the Rich Diagram.

This project is designed for the company “White Lotus”, which offers yoga classes all over London. A successful yoga centre, which have studios in 4 different areas of London and they are going to open 3 more studios in the next two years. Most of the classes had waiting lists which were manually managed.

Each studio manager keeps a spreadsheet about the information regarding the classes at a particular studio. The company has a General Manager, Mr. Yellow, who communicates with the Studio Managers and reports directly to the Board of Directors. The workshops and courses are advertised by the General Manager via email to a list of charts who provided their details at the studio and also via fliers which are available at the studio.

Clients can reserve through a phone call, email or in person. For reservation, Managers checks if there are any spaces left before making it. A client can cancel reservation up to 24 hours prior to the class by calling the studio. When booking, client has to book all the classes for that course. The cost of a single course is the sum of all the total classes with a discount of 10% and if a client reserves multiple courses within the same period of time, the discount will be 20%.

The online payment will be done by a Credit Card. Checking is done by the company’s bank, where they use their own system to provide an interface between the company and their bank’s credit check services.

Studio Managers manage the waiting list. If all classes are booked, then a client want to book, is kept on the waiting list. If there is cancellation, then first person on the waiting list

is selected. They are contacted by the manager via SMS or email. Almost everything is done manually.

The General Manager wants the system to be able to give information about yoga, types of classes or workshops and their schedule. Also, to be able to reserve places for class and manage waiting list.

The Board of Directors want the system to provide a way for the customers to leave reviews for a particular class and allow customers to search and browse for classes, read description and reviews and write reviews.

There are people who work under Mr. Yellow who are not particularly keen on the development of the new system, as they would to continue with their old method of working, which is to deal with people rather than to deal will computers. They are also unenthusiastic to go through training.

Process

The model of the system has been designed to the organization could have all the required information for the management and the further improvement of the business. The organization can easily find out all the information about the students, instructions and regular classes of the yoga class.

There are two model that our team has looked in to which is the Waterfall Model which is a linear sequential life cycle model. It is very simple to understand and use. There are several phases in this model and each phase must be completed before the next phase is started. The other model is the Agile Model, the agile model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction.

The Waterfall Model

The waterfall model is the earliest SDLC (Software Development Life Cycle) approach that was used for software development. It is typically separated into different phases which are most commonly known as the Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. Each phase has their own outcome and each phase must be completed to go to the next one.

Advantages of Waterfall Model

This model is allowing for departmentalization and control, where a schedule can be set to meet the deadline at each phase of the development. Each phase will move in a strict order. Each phase is well defined and will be easy to manage due to the rigidity of the model.

Disadvantages of Waterfall Model

However, this model does not allow much reflection or revision. It will be very difficult to change something at a later stage, if something was not well documented. No working

model is created until late in the life cycle. This model will not produce a working model until late at stage, which creates high amount risks and uncertainty to the project.

The Agile Model

The agile model will handle things differently and that each project has their own methods and need the best methods to suit the project. An iterative approach is taken and a working software is produced after each iteration. Each build is having its own features and the final build holds all the features that are required by the customer.

Advantages of Agile Model

It is suitable for fixed or changing requirements. In this model the functionality can be build quickly and it can be shown to the customer. It has minimal rules, which accounts for easy documentation and therefore be easy to managed. It is good model for projects that are always steadily changing, as it gives the developer more flexibility to develop the system.

Disadvantages of Agile Model

This model will depend on the interaction with the customer, meaning if the customer is not clear about their requirement the management will be creating the wrong system. Due to minimum documentation, it may be difficulties to change teams as there is lack of documentation for the new to understand the concept of the new system.

The Selected Model

Our team has decided to select the Waterfall Model because:

This model has a linear sequential life cycle model and is very easy and simply to understand. This model will allow maximum documentation unlike the Agile Model, so if the current team were to be replace the transfer would smooth.

There are no ambiguous requirements for the client, and the requirement are very well understood, clear and fixed.

The advancement in the program does not need to be checked by the client during the process.

It is easier to arrange and assign work because the phases are well defined, so they are processed and completed one at a time.

Project

A database for a yoga class needs to be developed for this task. The database will be made with Microsoft Access. A table will be created with all the required attributes for developing the database. Then the created table will be normalized to avoid transitive dependencies between the given attributes.

An ERD needs to be developed representing all the tables and their attributes and determining the connections between them. The database will be tested with some SQL

queries as mentioned in the requirements. A system design needs to be created with object-oriented methodologies for more flexible interactions and making required changes in the code. A UML class diagram will be created that will represent the attributes and methods of the database.

Databases have been visited together relevant information and generate ideas for developing an efficient database for the given subject. After making the strategies for formulation and implementation and preparing budgets, a clear vision of the requirements has been created.

Product

The final product should allow dynamically manage information for different types of classes or workshops sessions for each studio. Allow registered members to make reservations. Automatically handle waiting lists and allow clients to read and write reviews about the yoga class. The system should also allow the customer to browse and search for specific yoga classes that they want to attend.

The organization will be able to manage their business by the following g information: the classes that taught by a specific instructor, the reviews given to that specific class, the number of people that attend the class and other relevant information. This will allow the organization to make any major or minor changes in how the business is being run .

People

- Customer

The people who want join and attend the yoga course. They will write reviews about the company and also view their webpage to search and browse about different types of course they can attend.

- Studio Manager

Studio Managers are the managers who manages a specific studio. They work under the General manager and have to report everything to the General manager. They have information about the classes and manages the waiting and reservation list.

- General Manager

The General Manager is Mr. Yellow, who communicates with the Studio Managers to find out how the studios are performing and reports it to the Board of Directors.

- Board of Directors

The Board of Directors are a group of people that oversee the whole business and the give the company a sense of direction to head into.

- Bank

The bank will use its own system to provide an interface between the company and the bank credit check services to check the validity of the credit card.

- Yoga Instructor

People who teach the customers yoga. They are the people that have direct interaction will the customer.

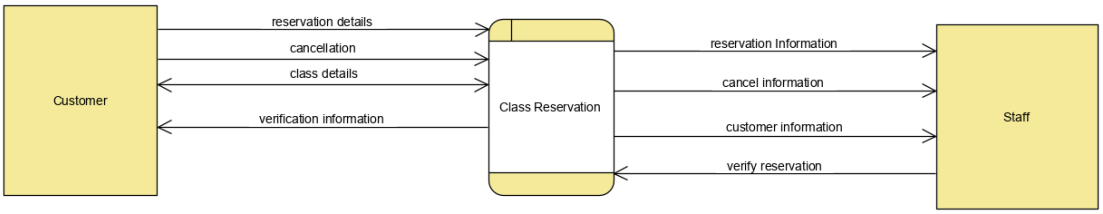
Functional Requirement

1. View the categories of the classes
2. To enter into the system the customer has to register. This will require the customer to fill in details.
3. Make reservation of the class. It will identify classes and manage reservation for that class.
4. Facilitate the customer to login to the system. Customer will have to enter username and password, to get the profile opened.
5. Search for specific classes in the system. Allows the customer to browse through the list of classes.
6. Allow the staff to sort the reservation and the waiting list.
7. Give the customer an opportunity to cancel their class.
8. Allow the customer to be able to read and write reviews about a specific class.
9. Allow customer to make payments by calculating the total amount that has to be paid.
10. Allow the system to check and verify the validity of the credit card using the banks credit card checking interface.

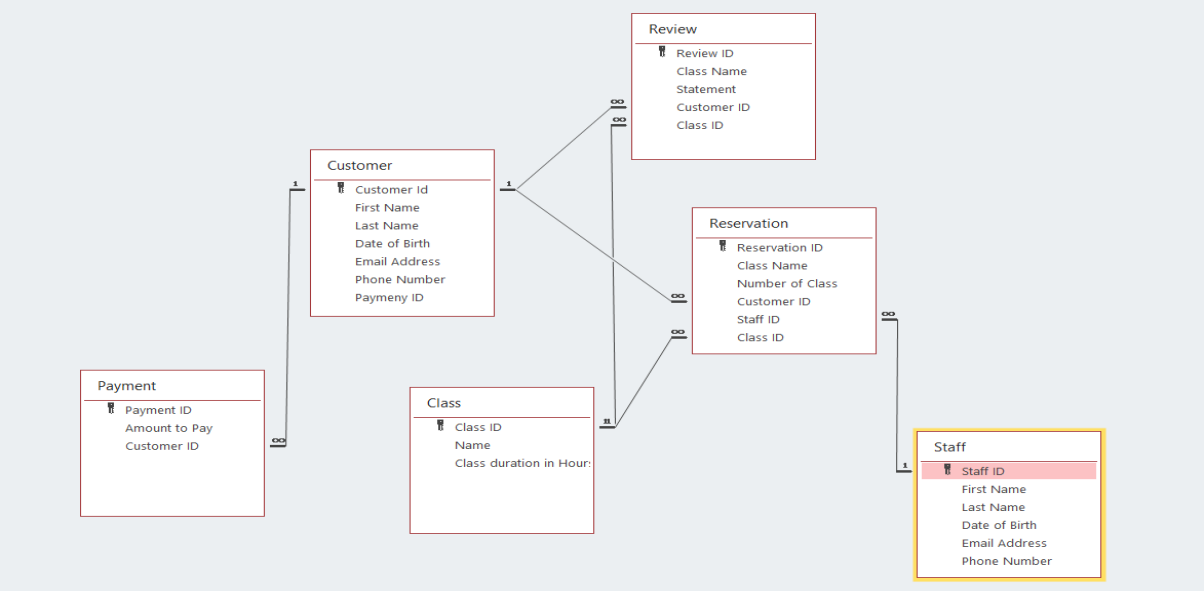
Non-Functional Requirement

1. Many customers can access the system at any time and maintain a maximum speed
2. The system server should be working 24 x 7
3. The system should have a strong security, because of the sensitive details the customer inputs into the system
4. The system should be reliable as to perform any queries that are desired by the customer
5. The system should be portable as to perform on different devices such as Personal Computers or Smart Phones.

Data Flow Diagram



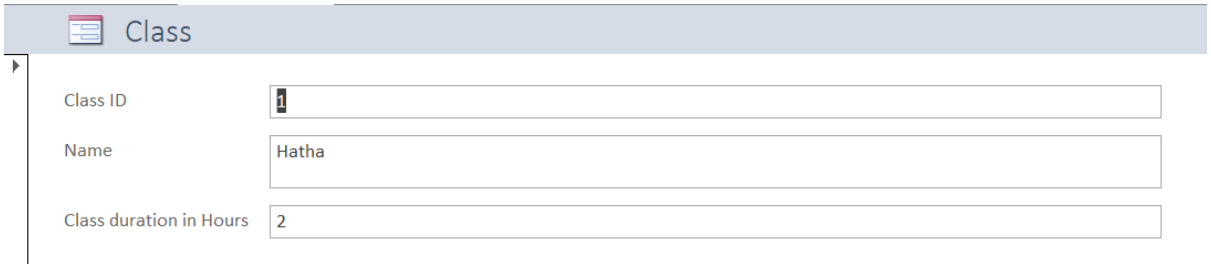
The ERD



These are the Following Test that were Carried Out

- Add a Class to the System.

This was done using a form.

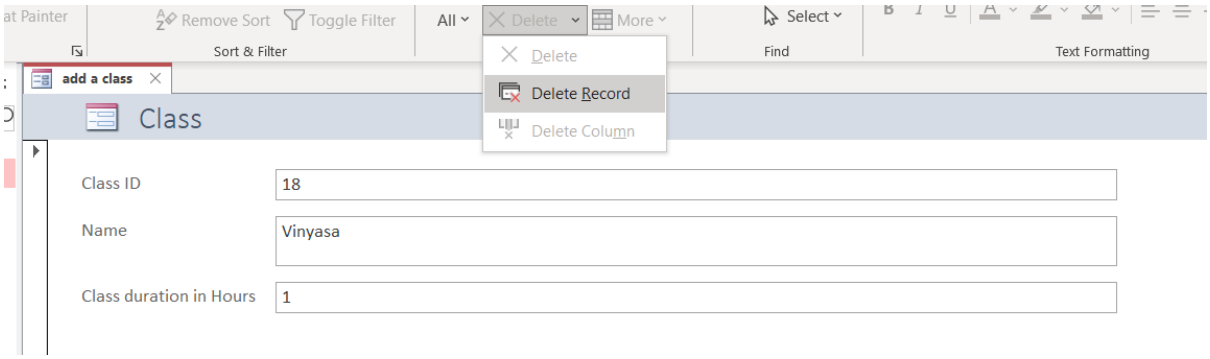


This was the outcome.

Class ID	Name	Class duration in H	Click to Add
1	Hatha	2	
2	Ashtanga	1	
3	Bikram	2	
4	Iyengar	3	
5	Kundlini	1	
6	Restorative	2	
7	Yin	2	
18	Vinyasa	1	
(New)		0	

- Delete a Class.

This was also done using a form.

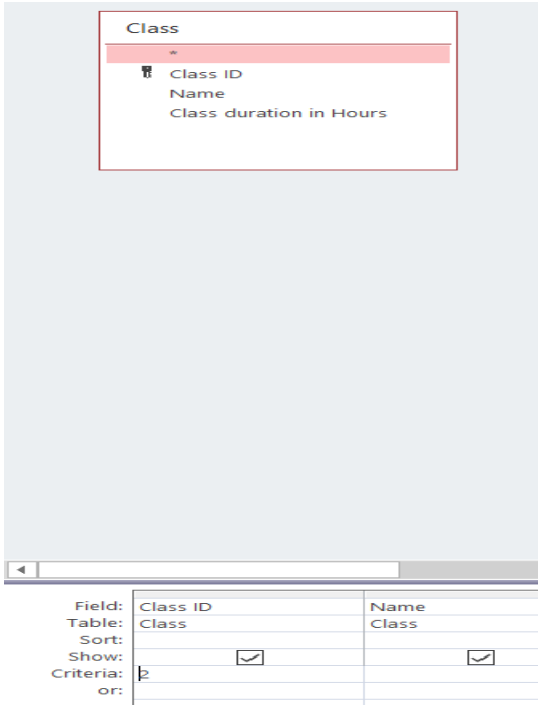


This was the outcome.

Class ID	Name	Class duration in H	Click to Add
1	Hatha	2	
2	Ashtanga	1	
3	Bikram	2	
4	Iyengar	3	
5	Kundlini	1	
6	Restorative	2	
7	Yin	2	
(New)		0	

- List of all Class of a Particular Type.

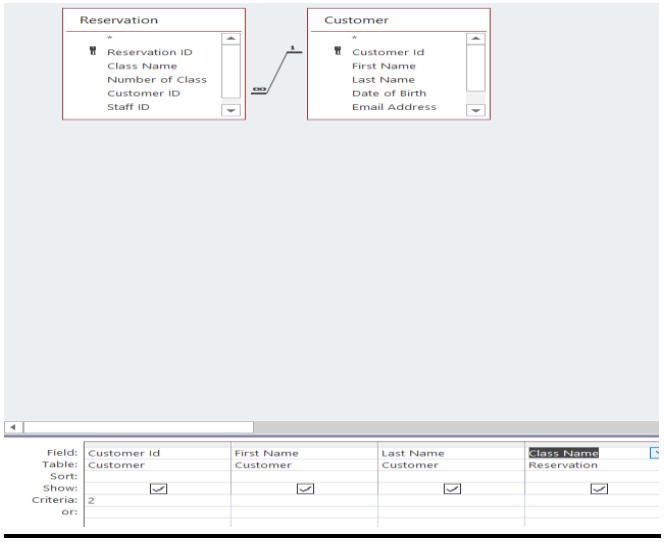
This was done using a simple query function.



This is the outcome.



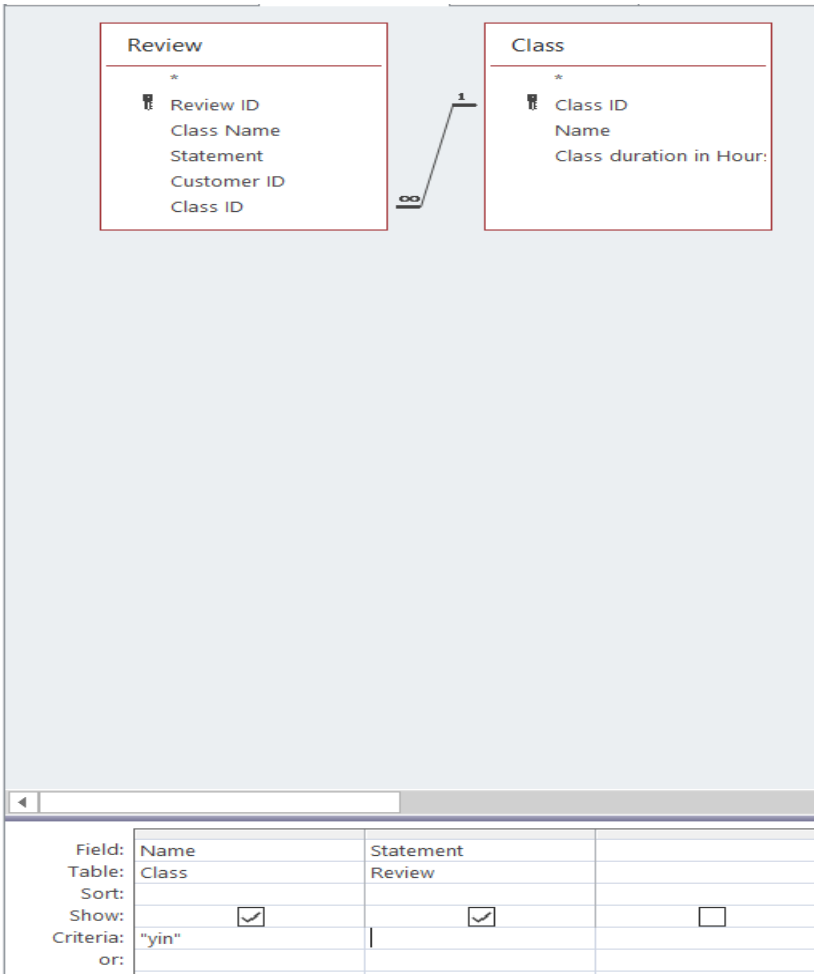
- Create booking for one class.



This is the outcome.

Customer Id	First Name	Last Name	Class Name
2	Smith	Harper	Ashtanga
2	Smith	Harper	Bikram
2	Smith	Harper	Iyengar
(New)			

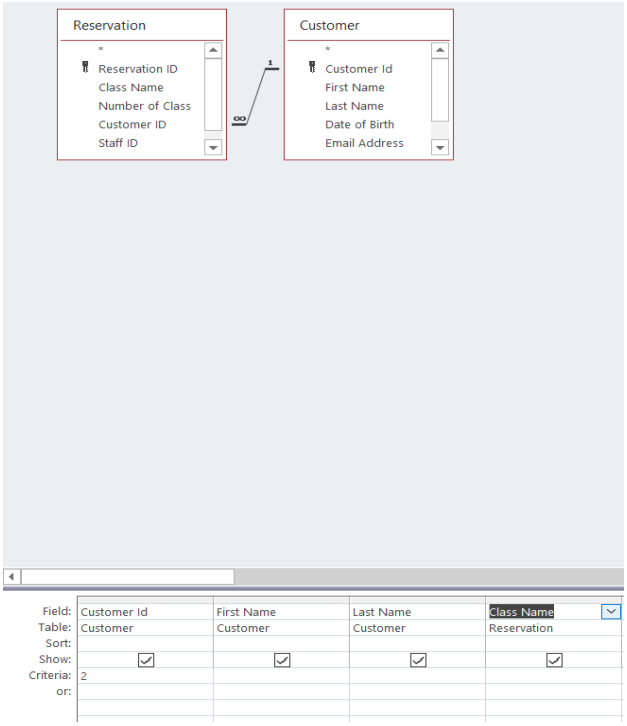
- Check all reviews for a particular yoga class.



This is the outcome.

Name	Statement
Yin	Very Good
Yin	Amazing

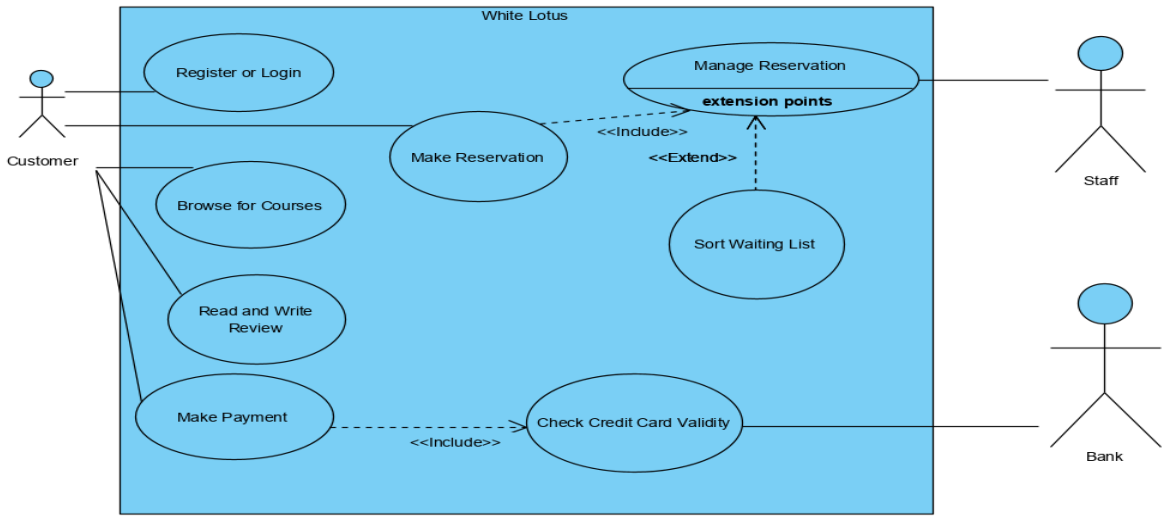
- Check all the bookings from a particular client.



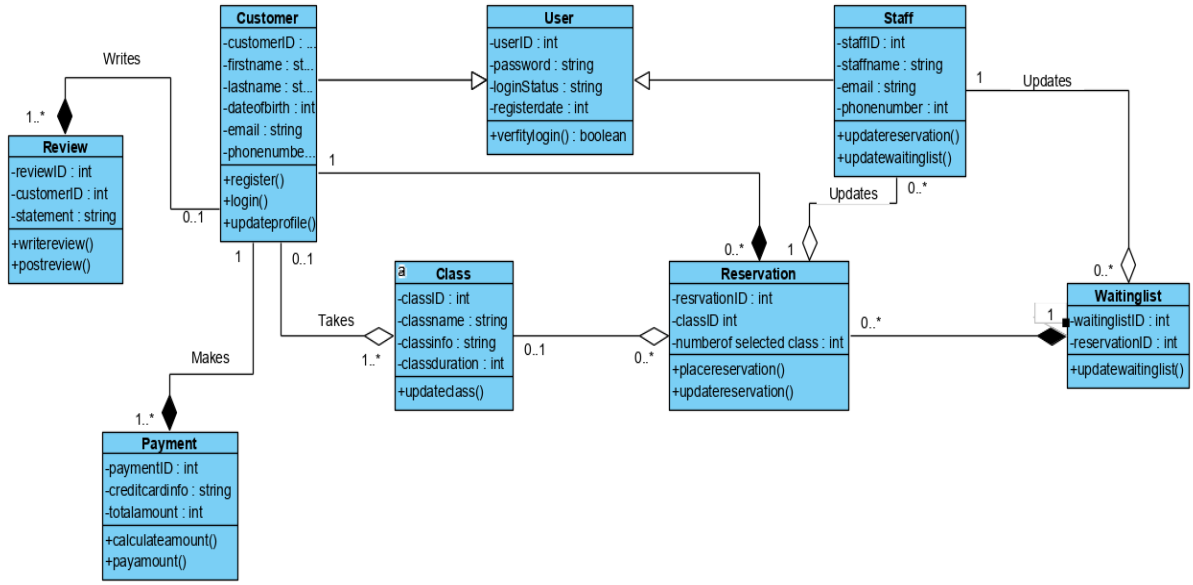
This is the outcome.

Customer Id	First Name	Last Name	Class Name
2	Smith	Harper	Ashtanga
2	Smith	Harper	Bikram
2	Smith	Harper	Iyengar
(New)			

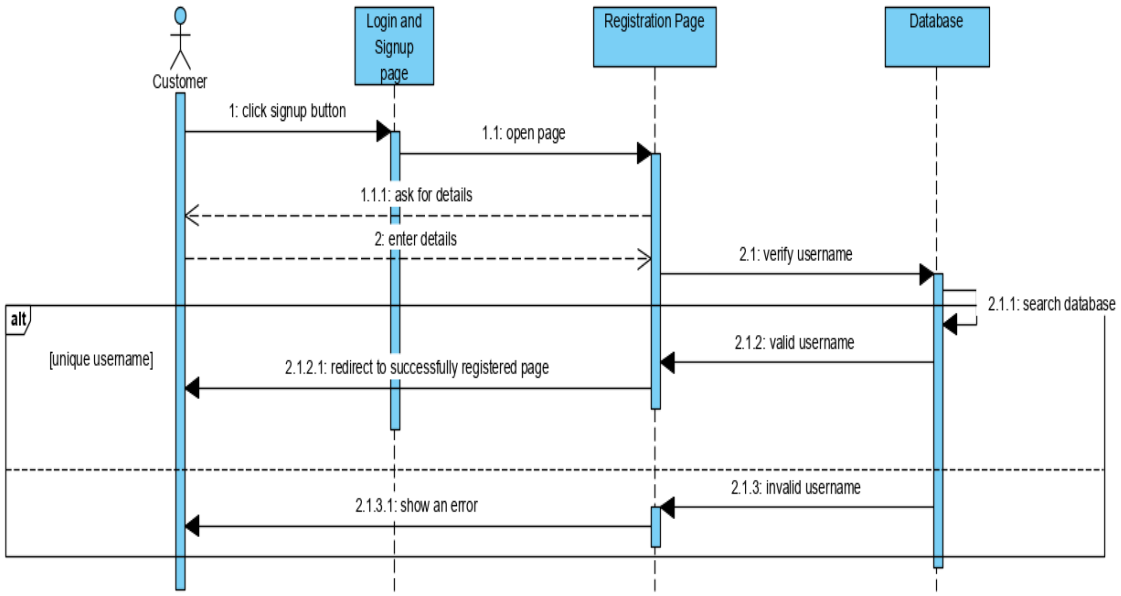
Use Case Diagram

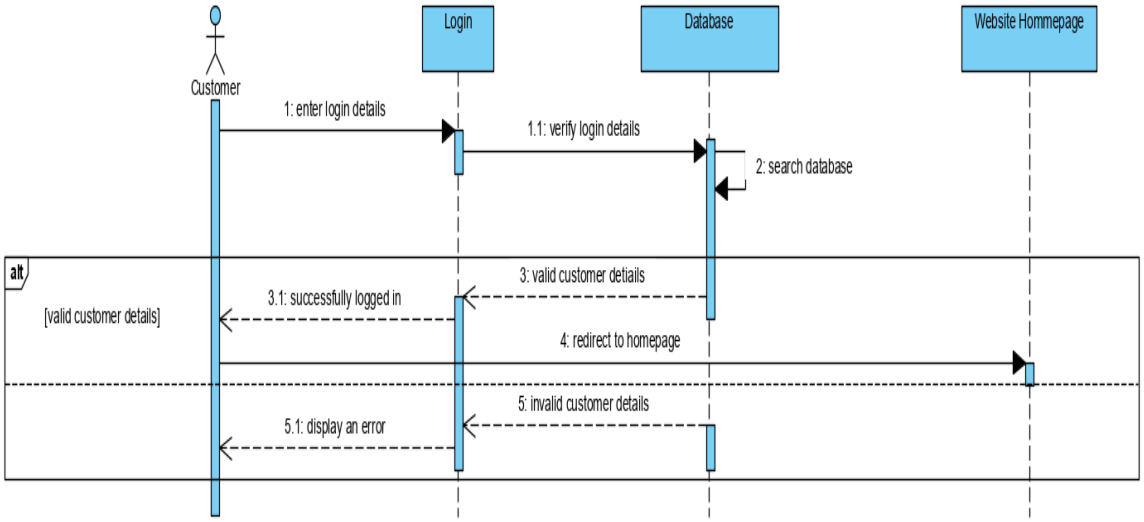


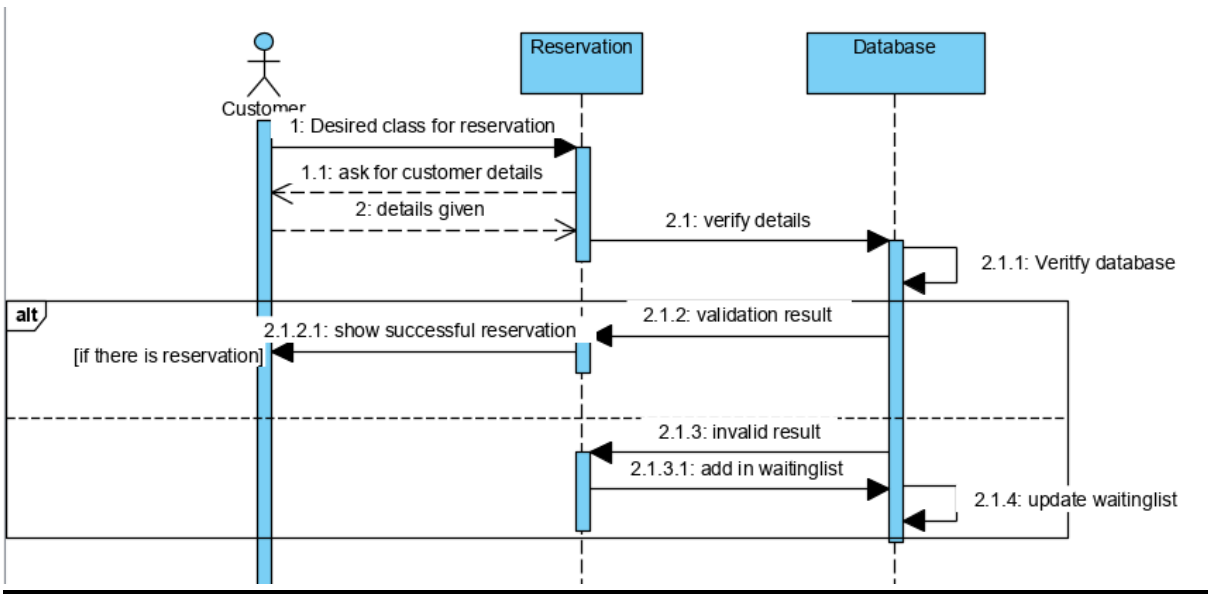
Class Diagram



Sequence Diagram







GRASP Pattern

Assign responsibilities so that the coupling remains low. The low coupling suggests class because class has to be coupled to reservation. There is also the reservation because the reservation has to be coupled to the waiting list as well. Add high cohesion, means that we have assign responsibilities. Giving responsibilities to class and reservation will support higher cohesion, as well as low coupling.

In this system the customer will be the controller as it gives account to class, reservation and payment.

The creator pattern suggest that the class is the creator as it has the responsibilities for the reservation and payment as well.

The staff will have all the instances of the number of classes, reservation and waiting list.

Conclusion

A database for a yoga class has been developed in this. First, some databases of yoga classes have been visited to gather relevant information and ideas. After gathering the information and analysing it properly, the required techniques of this job have been selected. Then the designing of the database has been started with creating a table with the selected attributes. The tables have been normalized to avoid transitive dependencies and after the normalization total of five tables have been created. The ERD has been representing all the attributes of each table and connections between the tables. After that, a system design of this database has been developed using object-oriented methodologies. A UML diagram based upon this design has been made.

Reference

SDLC-Waterfall Model *tutorials point*
https://www.tutorialspoint.com/sdlc/pdf/sdlc_waterfall_model.pdf

SDLC-Agile Model *tutorials point*
https://www.tutorialspoint.com/sdlc/pdf/sdlc_agile_model.pdf

Self-Assessment Sheet

Place **a tick in the box that you deem to be most indicative of the quality of the work** use the marks mapping below.

Marks mapping:

80-100 Exceptional	70-79 Excellent	60-69 Very Good	50-59 Good	40-49 Satisfactory	0-39 Fail
A	**B**	**C**	**D**	**E**	**F**

Please include comments for each section of the self-assessments. Also, please state the names of the students who completed that particular task

Note: You must submit this self-assessment together with the report. The boxes in bold are for the tutor.

	A	B	C	D	E	F	Comments
the 5 Ps • Problem: 3 marks		✓					This was the start of the project and which defines how the project was headed. This was particularly important phase which structured the whole project. The 5p's were relatively easier to complete compared to the others. This part was done collectively together in a group.
• Process: 10 marks			✓				
• Project: 4 marks			✓				
• Product: 3 marks			✓				

• People: 4 marks		✓					
functional and non-functional requirements: 5 marks			✓				In this we had understand what the customer wanted and how it had to be implemented Akshay and I had mostly worked on this together.
Structured design, • Entity Relationship Diagram 8 marks • Data Flow Diagrams. 5 marks • SQL queries 12 marks			✓	✓			This was much harder than the rest due to creating the database which took a lot of time. This was mostly done by Ayana and me.

UML Design Use Case Diagram - 8 marks Design Class diagrams - 9 marks Sequence Diagrams – 9 marks			✓	✓			This part was defined the system and how it had to be implemented. We had quite a bit of arguments in this phase. This part was done by all of us together.
Use of patterns – 5 points			✓				This was the most confusing part and took a lot of time just understanding the concept. This part was done by me only.

Acceptance testing (demo) - 10 marks							
Accurate Self-Assessment - 5 marks							
Possible AO?							

Group members work contribution form

In percentage, please indicate the work contribution of each member. This should be agreed by all group members**. The total of all members' work must add to 100%**

You must **submit this form in the final report.** Put your initials in the signature columns.

Each member must also bring a copy to the demonstration. This copy must be signed by all members.

Group/Team Name:_____

Team member name	Student ID	individual overall work contribution (%)	Signature
Student1: Tejesh Ramesh Bawa		40%	Tejesh
Student2: Akshay Suresh Kumar	001179739-8	30%	Akshay
Student3: Ayana Mangara Ashokan	001191154	30%	Ayana
Student4:			
Student5:			
		Total 100%	

Describe each task you performed (for example: creation of Use case diagram)	Student 1: work contribution in %	Student 2: work contribution in %	Student 3: work contribution in %	Student 4: work contribution in %	Student 5: work contribution in %	total
Task: The 5p's	40	30	20			100%
Task: Functional and Non-Functional Requirements	50	40	10			100%
Task: Structured Design	50	10	40			100%
Task: UML Design	36	32	32			100%
Task: GRASP Pattern	90	5	5			100%
Task:						100%